

Why do short interest levels predict stock returns?

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Abstract

High levels of short interest predict negative abnormal returns, but the reasons for this predictability are not well understood. Moreover, two popular explanations suggest very different interpretations. According to Miller (1977), stocks are overvalued in the presence of short-sale constraints and the subsequent negative abnormal returns represent a correction of this overvaluation. In contrast, Diamond and Verrecchia's (1987) information-based explanation does not presume that stocks are mispriced. Instead, short-sellers are regarded as traders who have information about future returns. We discriminate between the two explanations and find support for the information-based explanation. Therefore, our evidence suggests that short sellers incorporate value-relevant information into prices.

Keywords: Short selling, short interest, overvaluation, stock returns

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Abstract

High levels of short interest predict negative abnormal returns, but the reasons for this predictability are not well understood. Moreover, two popular explanations suggest very different interpretations. According to Miller (1977), stocks are overvalued in the presence of short-sale constraints and the subsequent negative abnormal returns represent a correction of this overvaluation. In contrast, Diamond and Verrecchia's (1987) information-based explanation does not presume that stocks are mispriced. Instead, short-sellers are regarded as traders who have information about future returns. We discriminate between the two explanations and find support for the information-based explanation. Therefore, our evidence suggests that short sellers incorporate value-relevant information into prices.

1. Introduction

A puzzling empirical finding is that stocks with high levels of short interest are followed by negative abnormal returns (Asquith and Meulbroek 1996, Desai et al. 2002). This finding appears to be inconsistent with the Efficient Markets Hypothesis because it implies that current public information can predict future stock returns. The literature offers two popular explanations for this predictability that differ substantially in their implications.

The first explanation, based on Miller's (1977) model, is that stocks with high levels of short interest are overvalued because pessimistic investors are unable to establish short positions, leaving only the optimists to participate in the pricing process. When shorting costs are high, market forces are unable to prevent overpricing in the amount of shorting costs. In this case, the future price correction would be directly proportional to shorting costs. The greater the shorting costs, the greater the possible overpricing; and therefore, the lower the subsequent stock returns. We will refer to this as the *overvaluation hypothesis*.

The second explanation rests on Diamond and Verrecchia's (1987) information-based view of shorting constraints. In a rational expectations model, they show that the effect of shorting constraints is anticipated by market participants. As a result, shorting constraints reduce the informational efficiency of prices but do not lead to overvaluation relative to publicly available information. Diamond and Verrecchia also point out that short sellers have no access to the proceeds from their sale because of collateral requirements. This precludes short selling for liquidity reasons, which, other things equal, should increase the fraction of short sellers who are informed. In brief, this view holds

that short sellers have a net informational advantage over other investors. We will refer to it as the *information hypothesis*.¹

We attempt to differentiate between the two hypotheses and find that the information effect is dominant. Our results suggest that short sellers are informed traders who can identify value-relevant information about publicly traded firms. In this capacity, short-sellers join the ranks of stock analysts, institutional investors, underwriters, auditors, and bank lenders, who have been shown to provide similar information-acquiring functions.²

The theoretical implications of the overvaluation and information hypotheses are strikingly different. In Miller's model, uninformed but irrationally optimistic investors take long positions while the irrationally pessimistic investors (equally uninformed) take short positions. When short positions are more difficult to establish than long positions, stocks will be overvalued as a result of the optimists' long trades. The overvaluation eventually dissipates when the fundamental value is revealed. Under the overvaluation

¹ A similar argument is made by Cohen, Diether and Malloy (2007) who contend that shorting demand signals informed trading. By differentiating between supply and demand shifts in the shorting market, Cohen, Diether, and Malloy show that stock returns are predicted only by changes in shorting demand, not shorting supply. Moreover, this predictability is stronger for stocks with less public information flow, as proxied by analyst coverage.

The information-based explanation is also consistent with practitioners' view of short sellers as investors with superior analytical skills (Gutfleish and Atzil, 2004). Under this explanation, investors short stocks when they can infer overvaluation from public sources. For example, Dechow et al. (2001) show that short sellers use information in fundamental-to-price ratios to take positions in stocks perceived to be overvalued, and unwind their positions as these ratios mean-revert.

² For example, underwriter quality has been shown to influence the extent of IPO underpricing (Beatty and Ritter (1986), Carter and Manaster (1990)), as has the identity of an IPO firm's auditors (Beatty (1989)). A long literature documents the positive effect of bank loan announcements on a firm's stock price (e.g., Mikkelsen and Partch (1986), James (1987), Lummer and McConnell (1989)), including the finding that announced loans from higher-quality lenders are associated with more positive borrower abnormal returns (Billett, Flannery, and Garfinkel (1995)). Also, Brennan and Subramanyam (1995) report that the equity of firms which are followed by a larger number of investment analysts trade with smaller bid-ask spreads, reflecting lower informational asymmetries across traders in the market. Finally, Boehmer and Kelly (2007) report that institutional investors improve the efficiency of security prices.

hypothesis, short sellers have no private information and therefore play no role in correcting this overvaluation.

In Diamond and Verrecchia's (1987) model, shorting constraints do not imply overvaluation relative to publicly available information. Instead, short sellers are regarded as informed traders whose absence from the market makes prices noisier. But market participants correctly anticipate the effect shorting constraints have on prices, and therefore stocks do not become systematically overvalued even in the presence of such constraints. Under the information hypothesis, short sellers play an active role in making prices more efficient by trading on negative information. Their positions, which are publicly observable, also provide long investors with a signal that the stock is overvalued. Subsequent actions from these long investors (selling or abstaining from buying) could accelerate the effect that short sellers have on prices. In contrast to the overvaluation hypothesis, this implies that the level of short interest is related to fundamental information about the stock.

Perhaps closest to our analysis is the work by Asquith, Pathak and Ritter (2005). They find that stocks with high levels of short interest and low institutional ownership subsequently earn abnormally low returns. To the extent that high short interest and low institutional holdings identify binding constraints on shorting, this result is consistent with the overvaluation hypothesis. But alternative interpretations are also possible. In particular, the higher shorting costs associated with lower institutional ownership is more likely to discourage uninformed traders than informed traders from shorting. As a consequence, more short selling in stocks with low institutional ownership could indicate that the marginal short seller is better informed. If true, the subsequent lower returns

reflect the adjustment to new information as in the information hypothesis, rather than the correction of mispricing as in the overvaluation hypothesis. Consistent with this interpretation, Asquith, Pathak and Ritter document that among short-sale constrained firms, “valuation” shorts have more predictive power than “arbitrage” shorts. Arbitrage shorts are not likely to reveal information about the stock, because they typically exploit the mispricing in related securities, such as convertible bonds issued by the same firm. This finding, while not fully exploited in Asquith et al.’s paper, suggests the information hypothesis could also play an important role in explaining their findings, and our analysis is intended to differentiate between the two explanations.

Prior evidence lends support to both hypotheses. Boehme, Danielsen, and Sorescu (2006) provide direct evidence in support of the overvaluation hypothesis. Using proxies for short sale constraints and dispersion of investors’ beliefs, they show that overvaluation is confined to those stocks that have both high short-sale constraints and high dispersion of beliefs. Diether, Malloy, and Scherbina (2002) find that dispersion of investors’ beliefs, as proxied by dispersion in analysts’ forecast, predicts lower subsequent returns. In the presence of short sale constraints, this predictability is consistent with the overvaluation hypothesis.

Several studies provide evidence that is consistent with the information hypothesis. Boehmer, Jones, and Zhang (2007) examine daily shorting flows and show that short sellers can systematically identify relatively overvalued stocks. Highly shorted stocks earn significantly lower risk-adjusted return over the next 90 days than lightly shorted stocks, suggesting that shorters are highly informed traders. Desai et al. (2002) show that the higher the short interest the greater its informativeness and argue that the

negative relation between short interest and subsequent stock returns is due to short sellers being informed investors. There is also some evidence that short sellers possess information about fundamentals. For example, Christophe, Ferri, and Angel (2004) find that negative earnings surprises are preceded by abnormal short selling. Francis, Venkatachalam, and Zhang (2006) show that short sellers are able to predict downward analyst forecast revisions, while Desai, Krishnamurthy, and Venkataraman (2006) find that short sellers are able to anticipate earnings restatements. Finally, Wu (2007) and Boehmer and Kelley (2007) provide direct evidence in support of Diamond and Verrecchia's (1987) model. They construct several measures of informational efficiency and find that daily shorting flows are positively related to these measures. This suggests that more shorting makes prices more efficient, as predicted by Diamond and Verrecchia.

While the extant empirical evidence is consistent with elements from both the overvaluation and information hypotheses, it is insufficient to discriminate between the two explanations. Observing negative returns following periods with high short interest, the most prominent empirical result, is consistent with both hypotheses. Stocks can be overpriced in the presence of binding short sale constraints and dispersion of investor opinion, and we would observe underperformance when prices subsequently revert to equilibrium. Informed short selling would also be associated with subsequent negative returns, but in this case it is the negative information short sellers have that drives prices down.

We propose two novel approaches to discriminate between the two hypotheses. First, Miller (1977) argues that as short sale constraints become more binding, fewer pessimists enter the market, resulting in higher market prices. This implies a positive

relation between changes in shorting constraints and contemporaneous overpricing. Following Asquith, Pathak, and Ritter, we consider high levels of short interest (which indicates high shorting demand) and low levels of institutional ownership (which indicates a low supply of lendable shares) as proxies for shorting constraints. Thus, under the overvaluation hypothesis, we expect abnormal returns to be positively correlated with contemporaneous changes in short interest and negatively correlated with contemporaneous changes in institutional holdings.

The information hypothesis also predicts a positive relation between returns and contemporaneous changes in short interest, but for a different reason. Here, the positive relation arises because smart short sellers become more active when prices are rising (see Brent et al. 1990, Dechow et al. 2001, Diether, Lee, and Werner 2007). But in contrast to Miller, the information hypothesis predicts that changes in institutional holdings are also positively related to contemporaneous abnormal returns. Holding the level of short interest constant, a decline in institutional holdings implies that short positions are now more likely to contain negative information about fundamentals. This is because lower institutional holdings increase the cost of shorting, so the marginal short seller is more informed (Diamond and Verrecchia, 1987). Rational investors who observe a decline in institutional ownership without a corresponding decline in short interest can then infer the negative information and cause a decline in market stock prices. These theoretical predictions are summarized in Table 1.

We find that the relation between stock returns and contemporaneous changes in institutional holdings is indeed positive, a result that is consistent with the information hypothesis and inconsistent with Miller's overvaluation hypothesis.

For our second approach, we reconfirm that short interest can predict unexpected releases of fundamental information and provide new evidence on how this predictive ability is related to short sale constraints. Miller's prediction arises from disagreement among investors – short sellers are more pessimistic than other traders, but have no private information about fundamentals. In contrast, the information hypothesis implies that short sellers are informed and have information about fundamentals that other traders do not have. Moreover, the information advantage of short sellers should be greater in stocks with lower levels of institutional holdings, because the higher shorting costs for these stocks imply that short transactions are more likely to be initiated by informed traders (Diamond and Verrecchia, 1987). Table 2 summarizes the theoretical predictions related to this second testing approach.

We measure unexpected fundamental information by looking at earnings surprises, the new information revealed around earnings announcements. We define earnings surprises in two different ways: as the standardized deviation of actual earnings from a time-series model and as the abnormal stock return around the earnings announcement date.

We find that the level of short interest predicts earning surprises – higher short interest is associated with more negative earnings surprises in the future. The predictive power of short interest decreases with the level of institutional ownership. To the extent that institutional ownership proxies for the supply of lendable shares (Asquith, Pathak and Ritter), this finding is consistent with Diamond and Verrecchia's (1987) argument that the level of short interest is most informative in stocks with higher shorting restrictions.

Overall, our results suggest that the return predictability of short interest cannot be fully explained with a Miller (1977) argument. Rather, the results are consistent with the idea that the level of short interest is informative regarding future firm performance and that short sellers provide value-relevant information about future stock fundamental value, especially for stocks with low institutional ownership. These results corroborate recent evidence that short sellers contribute in important ways to price discovery in financial markets and imply that restrictions on short selling could impose significant indirect costs on other market participants. Moreover, our findings add to a long debate about the source of the negative relationship between short interest and future returns. Specifically, our evidence suggests that this relationship arises because high short interest implies that the marginal short seller contributes more to price discovery, and not because of the inherent inefficiency associated with pessimists being driven out of the market.

The rest of the paper is organized as follows. In Section 2, we describe the data sources and methodology. We present empirical results in Section 3 and conclude in Section 4.

2. Data and methodology

Data on stock returns are from the Center for Research on Security Prices at University of Chicago (CRSP) for all common stocks listed on NYSE, Amex, and NASDAQ (share code 10 or 11), between January 1988 and December 2005. Monthly short interest, *Short*, is based on outstanding shorts reported by the exchanges, standardized by the number of shares outstanding. Data on institutional ownership are obtained from 13-F filings, available from Thomson Financial. We define institutional holdings, *IO*, as the sum of the holdings of all institutions for each stock in each quarter,

divided by the number of shares outstanding obtained from CRSP. Stocks that have available return data but no reported institutional holdings are assumed to have zero institutional ownership.

Size is the market value of equity calculated as the number of shares outstanding times the month-end share price. *B/M* is the ratio of book to market value of equity. Book value is computed as in Fama and French (2002) and measured at the most recent fiscal year-end that precedes the calculation date of market value by at least three months.³ We exclude firms with negative book values. The stock's momentum (*MOM*) is computed as its raw return over the previous 12 months.

The magnitude of quarterly earnings and their announcement dates are obtained from COMPUSTAT. We use two measures of earnings surprises: standardized unexpected earnings (*SUE*) and cumulative abnormal returns around the announcement (*CAR*).

Following Foster, Olsen, and Shevlin (1984), we define *SUE* in month *t* as

$$SUE_t = \frac{EPS_q - E[EPS_q]}{\sigma_t} \quad (1)$$

where *q* is the quarter, EPS_q are the most recent quarterly earnings per share, $E[EPS_q]$ are expected earnings per share, and σ_t is the standard deviation of unexpected earnings

³ Book value is defined as total assets (data 6) minus total liabilities (data 181) plus balance sheet deferred taxes and investment tax credit (data 35) minus the book value of preferred stock. Depending on data availability, the book value of preferred stock is based on liquidating value (data 10), redemption value (data 56), or carrying value (data 130), in order of preferences.

$(EPS_q - E[EPS_q])$ over the preceding eight quarters. To estimate expected earnings we use a seasonal random walk model as in Chan, Jegadeesh, and Lakonishok (1996):⁴

$$E[EPS_q] = \alpha + EPS_{q-4} \quad (2)$$

The second measure of earnings surprises is the average daily market-adjusted return around the earnings announcement date (CAR), computed as in Brown and Warner (1985):

$$CAR_i = \frac{1}{4} \sum_{j=-2}^{j=1} (r_{ij} - r_{mj}) \quad (3)$$

where r_{ij} is the stock i 's return on day j and r_{mj} is the return on the equally-weighted CRSP market index on day j . Day $j=0$ is the earnings announcement date. This measure captures the market's reaction to the earnings surprise.

We begin with a standard portfolio-sorting approach to test the ability of short interest to predict future returns and earnings surprises. Each month, we rank stocks on the basis of short interest, changes in short interest, institutional ownership, or changes in institutional ownership, and assign them to decile portfolios. For each portfolio we compute either contemporaneous returns (measured over the calendar month of portfolio formation) or subsequent returns (measured over the calendar month following portfolio formation). We use both equally weighted raw returns and abnormal returns based on a four-factor model. The latter are computed as the intercept of the following regression:

$$r_{pt} - r_{ft} = \alpha + \beta_1(r_{mt} - r_{ft}) + \beta_2SMB_t + \beta_3HML_t + \beta_4MOM_t + \varepsilon_{pt} \quad (4)$$

⁴ For robustness we also estimate expected earnings using a seasonal AR(1) model: $E[EPS_t] = EPS_{t-4} + \alpha + \beta(EPS_{t-1} - EPS_{t-5})$. The results (untabulated) are very similar to those obtained with the seasonal random walk model.

where $r_{pt} - r_{ft}$ is the portfolio return minus the return on one-month T-bill, $r_{mt} - r_{ft}$ is the market excess return, SMB is the size factor return, HML is the book-to-market factor return, and MOM is the momentum factor return. Factor returns are obtained from Kenneth French's web site at Dartmouth College. Aside from the sorts, we also use a multivariate predictive regression to verify the robustness of our results.

3. Results

We begin by replicating Asquith, Pathak and Ritter's (2005) findings and report the results in Table 3. Panel A of Table 3 shows that heavily shorted stocks underperform lightly shorted stocks by 1.51 percent per month on an absolute basis, and by 1.72 percent on a risk-adjusted basis. For the most heavily shorted stocks (decile 10), the average short interest is 8.8 percent and the average monthly risk adjusted return is -0.52 percent ($t=-3.03$). The results presented in Panel A are quite similar to those obtained by Asquith Pathak and Ritter. However, as previously noted, these results are consistent with both the overvaluation and information hypotheses.

Panel B examines the role of institutional ownership (IO) for the most heavily shorted decile. Asquith, Pathak and Ritter (2005) find that low IO predicts negative returns for these stocks. Columns (1) and (2) in Panel B present the average level of IO and short interest for three portfolios sorted on IO. Columns (3) and (4) replicate the findings of Asquith, Pathak and Ritter: both the absolute (Ret) and abnormal (α) returns are lowest for the lowest IO group (IO1). Moreover, the difference in returns between the lowest and highest IO groups (IO3-IO1) is always significant at better than the ten percent level. While Asquith, Pathak and Ritter interpret this evidence as consistent with Miller's (1977) theory, our new evidence in columns (5) and (6) illustrates the relevance

of the information hypothesis. We report the average earning surprises (SUE and CAR) for each IO group, corresponding to the earning announcement immediately following the portfolio formation period (month t). We observe that low-IO stocks forecast not only negative future returns, but also negative future earning surprises. If shorting is more difficult for low IO stocks, this suggests that shorts in the IO1 portfolio are particularly informed about deteriorating fundamentals, a result consistent with Diamond and Verrecchia's information hypothesis.

We now turn to more formal tests designed to discriminate between the overvaluation and information hypotheses.

3.1. Changes in shorting constraints and contemporaneous returns

We now perform our first set of tests to discriminate between Miller's overvaluation hypothesis and the information hypotheses. Miller predicts that stocks with binding short sale constraints (high short interest and low institutional ownership) are overpriced. Since the extent of overpricing is proportional to the level of short sale constraints, it follows that overpricing should exacerbate as short sale constraints become more binding. Therefore, we should observe a positive relation between changes in short sale constraints and contemporaneous stock returns.

Asquith, Pathak and Ritter (2005), propose that stocks are short-sale constrained when they have low levels of institutional ownership (low supply) and high level of short interest (high demand). Thus, a decrease in institutional ownership or an increase in short interest (or both) would imply greater constraints on shorting. We examine each of these two dimensions separately.

Each quarter q , we sort stocks into quintile portfolios based on the level of short interest at the end of that quarter, and further sort each portfolio into quintiles based on changes in institutional ownership between the end of quarters q and $q+1$. We compute contemporaneous raw and abnormal portfolio returns from the end of quarter q to the end of quarter $q+1$. Under Asquith, Pathak and Ritter's interpretation, an increase in institutional ownership would imply an increase in the supply of lendable stocks, which in turn would relax short sale constraints. The overvaluation hypothesis predicts that relaxing short sale constraints should allow prices to come down to their fundamental value. Thus, if the overvaluation hypothesis holds, we expect a negative relation between stock returns and contemporaneous changes in institutional ownership.

The results are presented in Table 4. Panel A of Table 4 shows results obtained using raw returns. Contrary to Miller's overvaluation predictions, stocks with increasing levels of institutional ownership are associated with higher contemporaneous returns. To test the statistical significance of this relation, we employ hedge portfolios that take long positions in stocks with the largest increases in institutional ownership (*ΔIO5*) and short positions in stocks with smallest increases in institutional ownership (*ΔIO1*).

Across all short interest quintiles, large increases in institutional ownership (*ΔIO5*) are always associated with higher returns when compared to small increases (*ΔIO1*). The returns of the hedge portfolios are always positive and statistically significant at better than the one percent level. This is particularly true for the most heavily shorted stocks (*Short5*), where short sale constraints are presumably most binding: the return of the hedge portfolio is 7.71 percent per quarter ($t=15.23$). The overvaluation hypothesis would predict a negative hedge return. Not only is the hedge

return positive, but its magnitude is highest among all hedge returns obtained with various portfolios sorted on short interest. In fact, hedge returns are monotonically increasing with the level of short interest. Miller's overvaluation hypothesis would predict a monotonically decreasing relation.

Panel B of Table 4 repeats the analysis of Panel A with abnormal returns computed as the intercept from the four-factor Fama-French-Carhart model. The results are similar to those reported in Panel A, suggesting that the relation between contemporaneous returns and changes in institutional ownership is not driven by the common four factors. Overall, the results presented in Table 4 do not support the joint hypothesis that (1) Miller's model is correct and (2) institutional ownership proxies for the supply of lendable shares.⁵

3.2. The information content of short interest

In our second set of tests to discriminate between the overvaluation and information hypotheses, we ask if short interest can predict future changes in firm fundamentals. We use subsequent earnings announcements as proxies of value-relevant news events. Under the information hypothesis, if higher levels of short interest are associated with higher shorting costs (D'Avolio 2002; Boehme, Danielsen and Sorescu 2006), the predictive power of short interest should be increasing in the level of short interest. In addition, if institutional ownership proxies for supply of shortable stock

⁵ For completeness, we also examine the relation between contemporaneous returns and changes in the level of short interest. As previously discussed (cf. Table 1), this test cannot differentiate between the Miller and information hypotheses because a positive relation is expected in both cases. We rank stocks into quintiles based on institutional ownership and then rank each portfolio into additional quintiles based on changes in short interest. The results (not tabulated) show that changes in short interest are positively related to contemporaneous return, as expected.

(Asquith, Pathak and Ritter 2005), predictability should be inversely related to the level of institutional ownership (Diamond and Verrechia 1987). In contrast, the overvaluation hypothesis implies no relation between short interest, institutional ownership and subsequent earning surprises. The results, presented in Tables 5 and 6, provide strong support for the information hypothesis. Both tables address the ability of short interest to predict future unexpected earnings but differ in the econometric approach. The analysis in Table 5 uses portfolio sorts, while Table 6 presents results based on Fama-Macbeth regressions.

3.2.1. Portfolio sorts

Panel A of Table 5 reports descriptive statistics about earnings surprises during our sample period. We sort stocks into quintiles based on standardized unexpected earnings (SUE). For each group, we report the average value of SUE, as well as the average cumulative abnormal returns (CARs) surrounding earning announcements. As expected, the two measures are highly correlated. Stocks with the most positive earnings surprises earn a CAR of 0.53 percent per day during the four-day announcement window. At the other extreme, CARs are -0.43 percent per day for stocks with the most negative earning surprises.

The next three panels show that the current level of short interest and institutional ownership can predict earning surprises.

In Panel B we sort stocks into quarterly short interest quintiles and compute the next quarter's earning surprises (SUE and CARs). These are then averaged intertemporally, across all quarters. We find that short interest can indeed predict future earning surprises. In the case of SUE, stocks with the highest short interest predict lower

future values of SUE than stocks with lower short interest, and the difference is significant at the ten percent level. The predictability is even more striking when CARs are used as a measure of earning surprises: a hedge portfolio that is long in lightly shorted stocks and short in heavily shorted stocks earns 0.31 percent per day during the four days surrounding earning announcements, and this return is statistically significant at better than the one percent level. These results provide *prima facie* evidence that short interest is quite informative about changes in fundamentals that are yet to be incorporated into stock prices.

Under the information hypothesis, if institutional ownership is correlated with the supply of lendable shares (Asquith, Pathak and Ritter), we would expect the relation between short interest and future earnings surprises to be stronger for lower levels of institutional ownership, since shorts are more likely to be informed in this case (Diamond and Verrecchia). Miller's overvaluation hypothesis makes no such prediction. We explore the role of institutional investors in Panels C and D of Table 5.

We first sort stocks into quarterly quintiles based on the level of institutional ownership, and then sort each portfolio into quintiles based on the level of short interest. We then compute next quarter's earning surprises for each of the 25 portfolios, and average these across quarters. Panel C reports the results obtained with the SUE measure of earning surprises. The results obtained with the CAR measure are reported in Panel D. Consistent with the information hypothesis, the predictive power of short interest decreases monotonically with institutional ownership. For example, in Panel D, among stocks with low institutional ownership (IO1), a hedge portfolio that takes long positions in heavily shorted stocks and short positions in lightly shorted stocks earns an abnormal

return of 0.70 percent per day ($t=7.13$) around earning announcements. This value drops to 0.04 percent ($t=0.35$) for stocks with high levels of institutional ownership (IO5), but hedge portfolio returns are significant at the five percent level for each of the other IO quintiles. The results obtained with SUE in Panel C are very similar.

Overall, the results in Table 5 provide evidence in favor of the information hypothesis in that they show that short interest predicts negative future earning surprises, and this prediction is stronger for firms with low levels of institutional ownership (cf. Table 2). Under Miller's hypothesis, earning surprises are unrelated to current levels of short interest and institutional ownership.

3.2.2. Fama-Macbeth regressions

The portfolio sorting method allows for non-monotonic relationships between the sorting variable and the analysis variable, but a possible shortcoming is that correlations with excluded variables could result in spurious relations. To address this concern, we now estimate quarterly Fama-Macbeth regressions of earnings surprises on the level of short interest and institutional ownership, controlling for size, book-to-market, and momentum. Since the predictive power of short interest is expected to be stronger for lower levels of institutional ownership, we add an interactive term which is the product of these two variables. The information hypothesis predicts a negative sign on the short interest variable and a positive sign on the interaction term of short interest and institutional ownership (cf. Table 2). By contrast, the overvaluation hypothesis predicts no relation between short interest, institutional ownership and future earning surprises.

Table 6 presents the results. In Panel A we transform all independent variables into decile ranks and then standardize these ranks to take values between zero and one.

This transformation makes the coefficient interpretation more intuitive and comparable across variables, in addition to minimizing the effect of outlier observations. For robustness, we present in Panel B the results based on the continuous values of the independent variables. In both panels, we run cross-sectional regressions in each quarter and compute time-series averages of the coefficient estimates, as in Nagel (2005). In columns (1) and (2) of each panel we use SUE as a proxy for earning surprises, and in columns (3) and (4) we use CARs.

Similar to the results in Table 5, we find that short interest predicts subsequent earnings announcement surprises, and this predictability decreases with institutional ownership. For example, in Panel A, the coefficient of short interest in the CAR equation (column (3)) is -0.659 percent (t -value -10.61), suggesting that when institutional ownership is low, high levels of short interest predict future negative earning surprises. However, considering the interaction term between short interest and institutional ownership, this effect is close to zero and positive ($-0.659 + 0.886 = +0.227$ percent) when institutional ownership is highest. A similar result is obtained when SUE is used instead of CAR as dependent variable (columns (1) and (2)). The results in Panel B are similar to those reported in Panel A.

Overall, the results obtained from Fama-McBeth regressions confirm the conclusions from the earlier portfolio sorts: the predictive power of short interest for future earnings surprises is a decreasing function of institutional ownership, consistent with the information hypothesis.

4. Conclusion

Asquith et al. (2005) suggest that stocks with high short interest and low institutional ownership are short-sale constrained. They argue that the return predictability associated with short interest is caused by high shorting costs, which prevent negative opinions to be impounded into stock prices as predicted by Miller (1977). We expand their analysis by considering a competing “information hypothesis” based on Diamond and Verrecchia (1987). Our results suggest that short sellers’ superior information, rather than a Miller-type correction of mispricing, is the main reason for the return predictability associated with short interest.

First, we show that decreasing institutional ownership (which makes shorting more difficult) does not exacerbate overpricing, as it should under the overvaluation hypothesis. Second, we demonstrate that short interest is not only informative about future returns but also about future changes in fundamentals. Moreover, this informativeness is greater when institutional ownership is lower. Our results are consistent with Diamond and Verrecchia (1987) interpretation that higher shorting costs make the marginal short seller more informed. As a result, short interest is more informative when shorting is more difficult. Taken together, our results suggest that short sellers are sophisticated investors who have value relevant information about firms and position themselves in stocks with deteriorating fundamentals.

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Table 1

Discriminating between the overvaluation and information hypotheses using contemporaneous stock returns

Hypothesis	Sign of the correlation between contemporaneous stock returns and	
	Changes in <i>Short Interest</i>	Changes in <i>Institutional Ownership</i>
Overvaluation	Positive	Negative
Information	Positive	Positive

Table 2

Discriminating between the overvaluation and information hypotheses using predictability of future earnings

Hypothesis	Sign of the correlation between future earning innovations and	
	Current levels of short interest	Interactive effect of Institutional Ownership
Overvaluation	Null	Null
Information	Negative	Positive

Table 3**Short interest, institutional ownership, and future returns**

The sample contains common stocks listed on the NYSE, Amex, and Nasdaq during the period from January 1988 to December 2005. Each month, we sort stocks into short interest deciles based on *Short*, the ratio of short interest to shares outstanding. Panel A shows the time-series means of firm characteristics, holding-period raw returns (*Ret*), and abnormal monthly returns (α) computed as the intercept from a four-factor model including *mktrf*, *smb*, *hml*, and *umd*. All returns are measured over a one month horizon, during the month following portfolio formation. *IO* is the percentage of shares owned by institutions as reported in 13F filings. *Size* is the market value of equity and defined as share price times the number of shares outstanding. *BM* is the book value of equity divided by the market value of equity calculated at least three months before the short interest data. In Panel B, stocks in the highest short interest decile are further sorted into three groups based on the level of institutional ownership. For each group, we show raw returns, abnormal returns, institutional ownership, level of short interest as well as two measures of earning surprises: SUE and CAR. SUE is the change in earnings per share from quarter $q-4$ to quarter q divided by the standard deviation of unexpected earnings over the last eight quarters. CAR is the average market adjusted abnormal return (in percent) during the $[-2, +1]$ window around the earnings announcement date. T-statistics are based on robust standard errors. *, **, and *** denote significance level at the 10%, 5%, and 1% level, respectively.

Panel A: Sorts on short interest

	Short	IO	Size	BM	Ret	α
P1 (low)	0.004	12.79	66.63	1.09	1.99***	1.20***
	0.025	16.62	122.93	0.96	1.77***	0.97***
	0.090	20.78	730.98	0.91	1.76***	0.85***
	0.214	26.18	1,313.95	0.85	1.71***	0.77***
	0.402	31.85	2,102.61	0.78	1.46***	0.52***
	0.665	37.52	3,189.79	0.70	1.38***	0.39**
	1.052	42.40	3,065.11	0.67	1.14***	0.10
	1.685	45.91	2,465.56	0.65	1.09***	0.03
	2.968	46.82	1,792.51	0.69	0.77*	-0.23
P10 (high)	8.823	49.03	1,025.90	0.79	0.48	-0.52***
P1 – P10					1.51***	1.72***
<i>t-value</i>					4.36	8.14

Panel B: Sorts on institutional ownership for the most shorted decile

	(1)	(2)	(3)	(4)	(5)	(6)
	IO	Short	Ret	α	SUE	CAR
IO1 (low)	19.09	8.62	-0.28	-1.00**	-0.185***	-0.091***
	49.53	8.48	0.67	-0.31**	-0.039	-0.044
IO3 (high)	78.39	9.36	1.04**	-0.24	0.263***	0.108***
3 – 1	59.30	0.74	1.32***	0.76*	0.448***	0.199***
<i>t-value</i>			2.71	1.84	9.21	5.30

Table 4

Changes in institutional ownership and contemporaneous returns

The sample contains common stocks listed on the NYSE, Amex, and Nasdaq during the period from January 1988 to December 2005. In Panel A we first sort stocks into quintile portfolios based on short interest (*Short*) in quarter q . Then we sort each quintile into quintile portfolios based on ΔIO , the change in institutional ownership from quarter q to $q+1$. We compute average monthly returns over the same period (from q to $q+1$) and report time series means. In Panel B, we first sort stocks into quintile portfolios based on the percentage institutional ownership (*IO*) in quarter q . Then we sort each quintile into quintile portfolios based on $\Delta Short$, the change in short interest ratio from quarter q to $q+1$. We again compute average monthly returns over the same period (from q to $q+1$) and report time series means. Panel A presents results based on absolute (raw) returns and Panel B presents results based on abnormal returns, computed as the intercept from a four-factor model including *mktrf*, *smb*, *hml*, and *umd*. T-statistics are based on robust standard errors. *, **, and *** denote significance level at the 10%, 5%, and 1% level, respectively.

Panel A: Contemporaneous absolute returns and changes in institutional ownership
Short interest quintiles

	Short 1 (low)				Short 5 (high)
$\Delta IO1$ (low)	1.29***	0.56	-0.59	-1.33**	-3.04***
	1.08***	0.70**	0.39	-0.09	-0.69
	1.67***	0.78**	0.81**	0.67*	0.24
	2.13***	2.02***	2.07***	1.93***	1.90***
$\Delta IO5$ (high)	3.49***	4.09***	3.98***	4.10***	4.67***
5-1	2.20***	3.53***	4.57***	5.43***	7.71***
t-stat	8.82	9.52	11.48	10.63	15.23
All Stocks (Ret)	1.94	1.63	1.33	1.06	0.62
All Stocks (IO)	14.34	23.60	34.82	44.40	48.12

Panel B: Contemporaneous abnormal returns and changes in institutional ownership
Short interest quintiles

	Short 1 (low)				Short 5 (high)
$\Delta IO1$ (low)	0.41**	-0.37*	-1.57***	-2.30***	-3.14***
	0.29	-0.18	-0.49***	-1.01***	-1.69***
	0.94***	-0.03	-0.04	-0.19	-0.77***
	1.34***	1.20***	1.22***	1.07***	0.96***
$\Delta IO5$ (high)	2.64***	3.19***	3.04***	3.22***	3.75***
5-1	2.23	3.56	4.61	5.52	6.89
t-stat	8.05	8.36	10.97	9.94	12.84
All Stocks (Ret)	1.12	0.76	0.43	0.16	-0.39
All Stocks (IO)	14.34	23.60	34.82	44.40	48.12

Table 5

Quarterly short interest, institutional ownership, and earnings announcements surprises
The sample contains common stocks listed on the NYSE, Amex, and Nasdaq during the period from January 1988 to December 2005. In Panel A, we sort stocks into quintiles based on standardized unexpected earnings, (*SUE*), defined as the change in earnings per share from quarter *q*-4 to quarter *q* divided by the standard deviation of unexpected earnings over the last eight quarters. We report time-series means of *SUE* and announcement abnormal returns, CARs, defined as the average market adjusted abnormal return (in percent) during the [-2, +1] window around the earnings announcement date. In Panel B, we sort on quarter *q* short interest and report the time-series average of *SUE* and CAR. In Panels C and D we conduct an independent double sort, first on quarter *q* institutional ownership and then on quarter *q* short interest. Panel C reports the average *SUE* for the next-quarter earnings announcement, and Panel D reports the corresponding CAR. T-statistics are based on robust standard errors. *, **, and *** denote significance level at the 10%, 5%, and 1% level, respectively.

Panel A: Descriptive statistics on earning surprises

	SUE 1 (low)				SUE 5 (high)
SUE	-2.00	-0.22	0.14	0.60	2.19
CAR	-0.43	-0.19	0.22	0.42	0.53

Panel B: Earnings surprises and short interest

	Short1 (low)	Short2	Short3	Short4	Short5 (high)	1 – 5	t-value
SUE	0.14***	0.14***	0.21***	0.19***	0.05	0.09	1.94
CAR	0.29***	0.16***	0.07**	0.03**	-0.02	0.31	4.79

Panel C: SUE for the subsequent earnings announcement

	IO1				IO5	1 – 5	t-value
Short 1	0.14***	0.11***	0.08***	0.15***	0.24**	-0.07	-0.69
	0.12***	0.09***	0.07*	0.18***	0.32***	-0.20	-3.34
	0.09***	0.08**	0.12***	0.23***	0.32***	-0.23	-4.10
	0.11**	0.02	0.08*	0.18***	0.34***	-0.23	-3.85
Short 5	-0.10**	-0.11**	-0.16***	-0.01	0.25***	-0.35	-4.71
1 – 5	0.24	0.22	0.24	0.16	-0.01		
t-value	4.59	4.84	3.89	2.36	-0.53		

Panel D: CARs around subsequent earnings announcements

	IO1				IO5	1 – 5	t-value
Short 1	0.47***	0.31***	0.21***	0.15***	0.12	0.35	3.21
	0.33***	0.18***	0.13***	0.14***	0.06*	0.27	5.55
	0.27***	0.13***	0.05**	0.06***	0.07**	0.20	2.77
	0.02	0.05	0.01	0.02	0.07**	-0.05	-0.56
Short 5	-0.23**	-0.18***	-0.09**	-0.01	0.08**	-0.31	-3.62
1 – 5	0.70	0.49	0.29	0.16	0.04		
t-value	7.13	5.21	3.57	2.17	0.35		

Table 6

Regression analysis of the effect of short interest on future returns and future earnings surprises

The sample contains common stocks listed on the NYSE, Amex, and Nasdaq during the period from January 1988 to December 2005. We estimate quarterly regressions of excess return and earnings surprises on transformations of firm characteristics. Excess returns ($Ret-Rf$) are the average monthly excess stock return over the quarter after subtracting the average return on one-month T-Bill. Earnings surprises are measured using either SUE or CAR. SUE is the change in quarterly earnings per share from quarter $q-4$ to quarter q , divided by the standard deviation of unexpected earnings over the last eight quarters. CAR is the average market adjusted abnormal return during a $[-2, +1]$ window around the earnings announcement date. IO is the percentage of shares owned by institutions as reported in 13F filings. $Size$ is the natural logarithm of the market value of equity, which is defined as share price times the number of shares outstanding. BM is the natural logarithm of the ratio between book value of equity and the market value of equity calculated at least three months before the short interest data. MOM is the mean return performance over the past 12 months. In Panel A we transform all independent variables into decile ranks and standardize them to take values between zero and one. Panel B shows results using the actual values of the independent variables. Both panels report time-series averages of the coefficient estimates. T-statistics (in parentheses) are based on robust standard errors.

Panel A: Regressions based on standardized values of the independent variables

	(1)	(2)	(3)	(4)
	SUE	SUE	CAR	CAR
Intercept	0.113 (4.83)	-0.046 (-1.64)	0.455 (9.17)	0.253 (5.15)
Size	0.157 (1.82)	0.230 (2.42)	-0.402 (-6.31)	-0.329 (-5.05)
Bm	-0.695 (-11.73)	-0.678 (-12.20)	0.110 (2.55)	0.131 (3.05)
MOM	0.992 (17.50)	0.993 (17.72)	0.111 (3.48)	0.121 (3.74)
Short	-0.547 (-8.75)	-0.372 (-7.69)	-0.659 (-10.61)	-0.163 (-4.63)
IO	-0.062 (-1.02)		-0.337 (-5.79)	
Short*IO	0.314 (5.28)		0.886 (11.09)	
R ²	8.13%	7.91%	1.05%	0.81%

Table 6 (continued)

Panel B: Regressions based on actual values of the independent variables

	(1)	(2)	(3)	(4)
	SUE	SUE	CAR	CAR
Intercept	-0.296 (-3.04)	-0.333 (-2.77)	0.946 (11.31)	0.855 (9.99)
Size	0.028 (3.08)	0.031 (2.74)	-0.067 (-9.49)	-0.057 (-7.98)
Bm	-0.207 (-10.17)	-0.201 (-10.12)	0.044 (2.97)	0.052 (3.54)
MOM	6.467 (21.23)	6.616 (21.14)	0.728 (2.64)	0.855 (3.05)
Short	-0.056 (-9.49)	-0.025 (-7.55)	-0.043 (-10.62)	-0.014 (-4.95)
IO	-0.005 (-0.09)		0.065 (1.59)	
Short*IO	0.064 (5.91)		0.054 (8.47)	
R ²	6.82%	6.52%	0.93%	0.82%